IoT Based Home Security System

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1. Abstract

The proposed **IoT Based Home Security System** is a prototype designed to provide real-time monitoring and protection of a model residential building using multiple sensors, visual indicators, local alarms, and camera surveillance.

The system uses an **Arduino Nano** as the primary controller for monitoring security and safety sensors installed in three rooms. A **magnetic door sensor** detects unauthorized door opening, while **IR line-of-sight sensors** detect disturbances or movement in protected areas. A **flame sensor** provides early detection of a possible fire condition. Three LEDs represent the security status of three individual rooms, and an **OLED display** provides the user with information about the detected condition.

When an abnormal condition is detected, the Arduino activates a **vibration motor as a local alert** and displays the corresponding warning on the OLED. The system also monitors the availability of the main electrical supply. During a power failure, it detects the interruption and automatically switches the system to a **backup battery supply**, allowing the security system to continue operating.

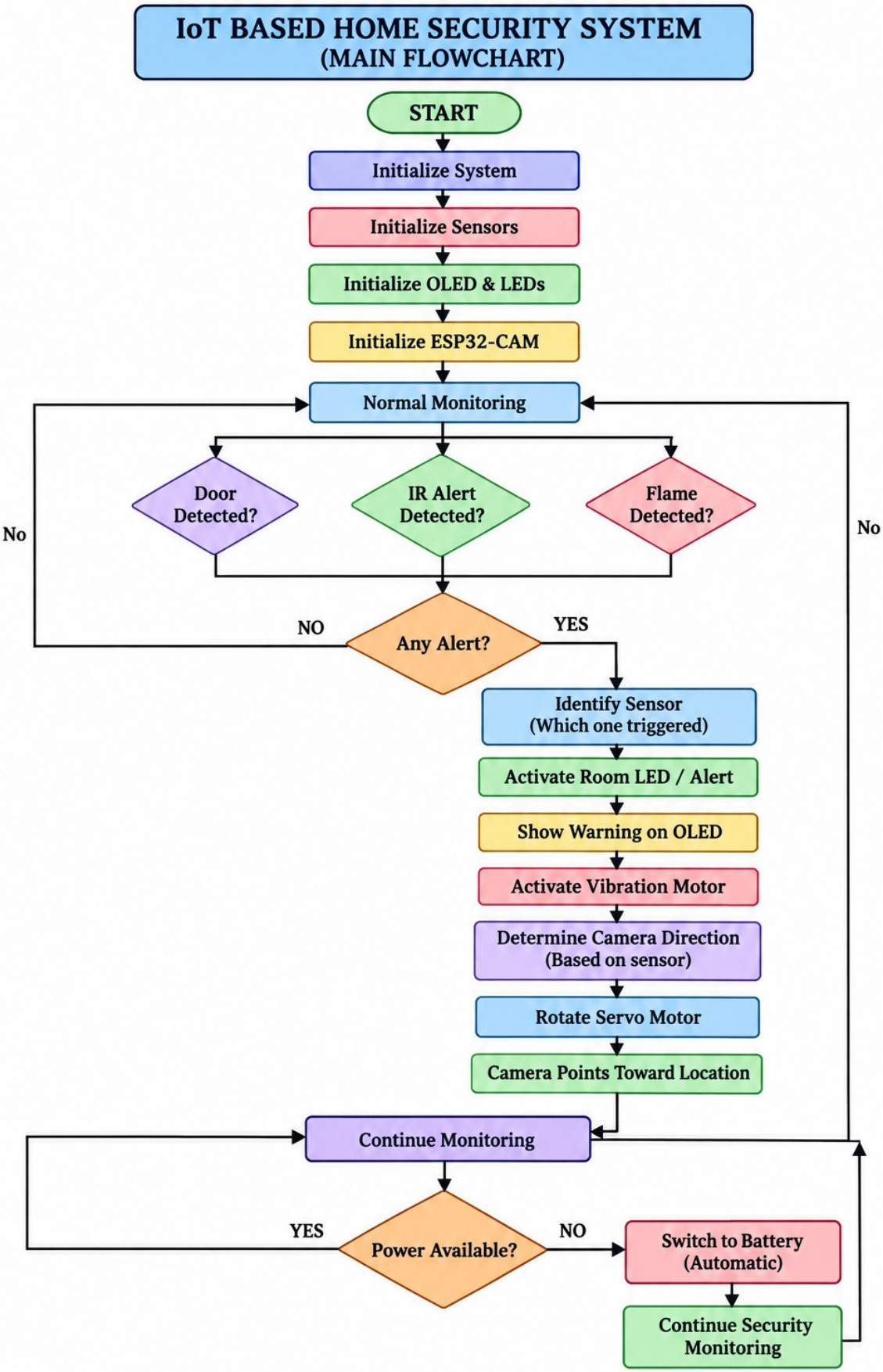
A separate **ESP32-CAM** provides wireless video surveillance. It operates as a Wi-Fi access point, allowing a user to connect directly to the camera and view the monitored area through a web interface satisfying the Internet of Things (IOT) requirement. The ESP32-CAM is mounted on a **servo motor**, allowing the camera to rotate toward the location where a security sensor has been triggered.

The proposed system therefore combines **intrusion detection, fire detection, power-failure protection, local alerting, and wireless video surveillance** into a single home-security prototype.

2. Incorporated Functions

- Door monitoring
- IR disturbance detection
- Flame detection
- Room status indication
- OLED information display
- Vibration alert
- Power-failure detection
- Battery backup control
- Determining which sensor has been triggered
- Sending the required direction information to the camera system
- Servo-controlled camera positioning
- IOT using ESP32 Cam Module

3. Flowchart


IoT BASED HOME SECURITY SYSTEM
(MAIN FLOWCHART)
START
Initialize System
Initialize Sensors
Initialize OLED & LEDs
Initialize ESP32-CAM
Normal Monitoring
Door Detected?
IR Alert Detected?
Flame Detected?
No
No
NO
Any Alert?
YES
Identify Sensor
(Which one triggered)
Activate Room LED / Alert
Show Warning on OLED
Activate Vibration Motor
Determine Camera Direction
(Based on sensor)
Rotate Servo Motor
Camera Points Toward Location
Continue Monitoring
YES
Power Available?
NO
Switch to Battery
(Automatic)
Continue Security
Monitoring

4. Scope of the Project

1. Intrusion Detection

The system can detect:

- Unauthorized door opening
- Disturbance/interruption of IR beams
- Activity in protected areas

2. Fire/Safety Monitoring

The flame sensor can detect the presence of a flame and immediately generate an alert.

The system can:

- Display a fire warning
- Activate the vibration alert
- Indicate the affected area
- Direct the camera toward the relevant location

3. Visual Surveillance

The ESP32-CAM provides:

- Wireless camera access
- Live video viewing
- Remote observation through a Wi-Fi-enabled device and monitor notifications
- Automatic camera positioning

4. Automatic Camera Positioning

The servo-mounted camera can automatically change its viewing direction based on the sensor that triggered the alarm.

5. Local User Notification

The system provides immediate local feedback using:

- OLED display
- Room LEDs
- Vibration motor

6. Power-Failure Protection

The system can detect main power failure and automatically change to backup battery operation. This ensures that the security monitoring system continues operating during a power interruption.

7. IoT Capability

The ESP32-CAM provides a wireless interface that allows the user to access the camera without requiring an external router, depending on the final implementation